



Naveen S Das

Master of Technology | Quantum Technology
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SUMMARY

M.Tech scholar in **Quantum Technology (IIT Jodhpur)** specialising in **Quantum Optimization and Quantum Machine Learning**, with hands-on experience in QAOA, Quantum algorithms, QUBO/Ising formulations, and variational quantum algorithms. Combines a strong foundation in Data science, Artificial Intelligence/Machine Learning and production-grade software engineering .

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech QT	Indian Institute of Technology, Jodhpur	9.25	2026
PGP Data Science	Great Lakes Institute of Management, Gurgaon	—	2022
B.Tech, ECE	Model Engineering College, Kochi	6.59	2020
Senior Secondary	Vidyodaya School, Thevakkal	91%	2016
Secondary	Campion School, Kochi	9.4	2014

EXPERIENCE

• Gadgeon Smart Systems Pvt Limited 2022 – 2023

Job Title : Data Scientist, Kerala

- Built and deployed **production-grade ML pipelines** for predictive maintenance and industrial analytics, enabling real-time fault detection and proactive asset management in IoT-enabled manufacturing environments.
- Developed end-to-end **predictive analytics systems** using supervised and time-series models (LSTM, Random Forest, XGBoost) on sensor data streams, reducing unplanned downtime and improving operational efficiency.
- Designed scalable data preprocessing and feature engineering pipelines for high-frequency industrial sensor data, ensuring model robustness under noisy, real-time conditions.
- Collaborated with cross-functional engineering teams on end-to-end delivery of data-driven automation solutions.
- Mentored junior team members in ML best practices, fostering a high-performing engineering environment.

PROJECTS

• Quantum Optimization for Vehicle Routing Problem (QUBO, Ising Models, QAOA & Quantum Annealing) 2025

Tools: Qiskit, PennyLane, Python, QUBO/Ising Mapping, OR-Tools, NISQ-Hardware, D-Wave Ocean SDK

- Formulated the **Vehicle Routing Problem (VRP)** and capacitated variants as QUBO/Ising models, enabling quantum-ready encoding for both gate-based (QAOA) and annealing-based solvers.
- Transformed QUBO into Ising Hamiltonian representation for compatibility with quantum optimization frameworks
- Analysed **QUBO-Ising mappings**, variational circuit design strategies, and parameter optimization methods (gradient-based and gradient-free) for real hardware constraints.
- Implemented Quantum Approximate Optimization Algorithm (QAOA) using Qiskit, pennylane and analyzed solution quality across parameter iterations
- Solved optimization problem using quantum annealing techniques on D-Wave Systems (Leap / Ocean SDK)
- Visualized optimized routes, energy landscapes, and convergence behavior using Python (NumPy, Matplotlib)

• Hybrid Quantum-Classical Neural Network for Image Classification

Ongoing

Tools: PennyLane, PyTorch, Qiskit, ResNet, Vision Transformers, Variational Quantum Circuits

- Designed a hybrid architecture embedding variational quantum circuits within classical deep learning backbones (ResNet / Vision Transformers) for classification tasks.
- **Optimised quantum feature maps and ansatz design** to maximise expressive power within qubit constraints of NISQ processors.

• BB84 Quantum Key Distribution Simulation with Decoy States

Tools: Python, Qiskit, Quantum Communication, QBER Analysis, Interactive Dashboard

- Simulated BB84 QKD with decoy-state extension using Qiskit; computed secure key rates and QBER under realistic channel noise.
- Built an interactive visualization dashboard for real-time protocol analysis and parameter tuning.

• Brain Computer Interface (BCI) – Canine Olfaction EEG Signal Classification

Tools: Python, MNE, Biosignal Processing, Neural Networks, Time Series Analysis, Neuroscience

- Built an end-to-end pipeline for EEG signal acquisition, preprocessing, signal analysis, feature extraction, and neural-network-based classification.
- Applied time-frequency analysis and data storytelling techniques for medical-grade signal interpretation.

• Retrieval-Augmented Generation (RAG) Application

Tools: GPT, Llama3, LangChain, Chroma & Pinecone Vector DB, FastAPI, Docker

- Architected and deployed a production-ready RAG system with vector-database retrieval, reducing hallucination rates through structured prompt engineering.
- Containerised with Docker and served via FastAPI

SKILLS

Quantum Optimization: QAOA and variants, QUBO/Ising formulations, Variational Quantum Eigensolver (VQE), Quantum Annealing, Combinatorial Optimization, Parameter Optimization (gradient-based & gradient-free), Noise Mitigation on NISQ devices

AI/ML & Deep Learning: PyTorch, TensorFlow, Scikit-Learn, Computer Vision (CNNs), NLP (LLMs, RAG), LangChain, Agentic AI, Vector Databases (Chroma/Pinecone), Model Deployment & Optimization

Programming & Data: Python, C++, SQL, Tableau, Data Analysis, NumPy, SciPy

MLOps & Cloud: Git, Docker, CI/CD Pipelines, FastAPI, Streamlit, MLflow, AWS, Google Cloud Platform (GCP)

Key Courses: Quantum Computation, Quantum Algorithms, Quantum Optimization, Quantum Machine Learning, Quantum Cryptography, Classical & Quantum Algorithms, Quantum Error Correction, Device Independent Quantum Technology

CERTIFICATIONS

- IBM Quantum Badges
- Computational Neuroscience – University of Washington (Coursera)

ACHIEVEMENTS

- Smart India Hackathon (SIH) 2025 – Finalist
 - GATE 2025 Qualified (Data Science & AI)
 - Business Quiz (BizQuiz) at IIT Jodhpur Techfest – Winner (2nd Place)
 - College and University level Football player
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